

INVESTMENT RISK ANALYSIS REPORT

84 Scenario Risk Model

Japan Property Investment

Based on 30 years of historical FX and property data, this report models 84 possible investment outcomes across 7 exchange rate scenarios, 4 property price trajectories, and 3 holding periods to provide a comprehensive risk assessment for Hong Kong investors.

Sample Property: Osaka Ten-no-Chaya Area

Investment: JPY 62,400,000 / HKD 3,200,000

Model: 7 FX x 4 Property Price x 3 Holding Periods

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1. Understanding Our Model

When you invest in Japanese property from Hong Kong, your final return depends on two fundamentally different types of factors. Our model separates all inputs into **Fixed Data** (known today) and **Variable Data** (unknown, scenario-tested). By testing all combinations systematically, we generate 84 distinct scenarios mapping the full range of possible outcomes.

Fixed Data (Known Today)

Variable Data (Unknown - Scenario Tested)

How the 84 scenarios work: We combine 7 exit exchange rates (JPY strengthening 33% to weakening 44%), 4 property price trajectories (-3% to +3% annual), and 3 holding periods (5, 7, 10 years) = 84 outcomes. Each outcome is measured in ROI (%) and absolute HKD gain/loss.

Why no management fee? This analysis assumes direct ownership where the investor handles property management. The only unavoidable costs are property tax and building insurance (0.3%/year). This significantly improves net cashflow compared to fund-based investments.

2. Your Investment Parameters

The property is located in the Ten-no-Chaya area of Osaka. Below are the fixed financial parameters forming the base of our 84-scenario model.

Property Purchase Price	JPY 62,400,000
Entry Exchange Rate	19.5 JPY/HKD
Property Value in HKD	HKD 320.0M
Loan Amount (40% LTV)	JPY 2,496 wan
Your Equity (Down Payment)	JPY 3,744 wan = HKD 192.00M
Mortgage Rate	3.0% p.a. (fixed)
Loan Term	15 years
Monthly Mortgage	JPY 172,369
Annual Mortgage	JPY 2.07M

Annual Cashflow Breakdown

Loan Balance at Exit

3. Market Variables: The Two Unknowns

3.1 Exchange Rate Scenarios (JPY/HKD)

The current JPY/HKD rate is ~19.5. Over 30 years, it ranged from 622 (1998, JPY very strong) to 1,181 (2024, JPY very weak). Our 7 scenarios cover JPY strengthening 33% (rate=13.0) to weakening 44% (rate=28.0).

3.2 Property Price Scenarios

Japan property has seen dramatic cycles: post-bubble decline (1990s-2000s), stagnation, then urban recovery (2012-present). We test 4 annualized rates from -3%/yr (1995-2005 national) to +3%/yr (2015-2025 Tokyo/Osaka trend).

4. 84 Scenario Results

Each cell shows ROI (%) and HKD gain/loss for your HKD 1.92M equity investment. Green = positive, yellow = modest, red = loss.

4.1 Holding: 5 Years

Prop \ FX	13.0	15.0	17.0	19.5	22.0	25.0	28.0
-3%/yr	+73% +140k	+50% +96k	+32% +62k	+15% +29k	+2% +4k	-10% -19k	-20% -38k
-1%/yr	+96% +184k	+70% +134k	+50% +96k	+31% +59k	+16% +30k	+2% +4k	-9% -17k
+1%/yr	+121% +232k	+92% +176k	+69% +133k	+47% +91k	+31% +59k	+15% +29k	+3% +5k
+3%/yr	+148% +284k	+115% +221k	+90% +172k	+65% +126k	+47% +89k	+29% +56k	+15% +29k

4.2 Holding: 7 Years

Prop \ FX	13.0	15.0	17.0	19.5	22.0	25.0	28.0
-3%/yr	+85% +163k	+60% +116k	+41% +79k	+23% +45k	+9% +18k	-4% -7k	-14% -27k
-1%/yr	+116% +222k	+87% +167k	+65% +125k	+44% +84k	+28% +53k	+12% +24k	+0% +0k
+1%/yr	+151% +290k	+117% +225k	+92% +176k	+67% +129k	+48% +93k	+30% +58k	+16% +32k
+3%/yr	+190% +365k	+152% +291k	+122% +234k	+94% +180k	+72% +137k	+51% +98k	+35% +67k

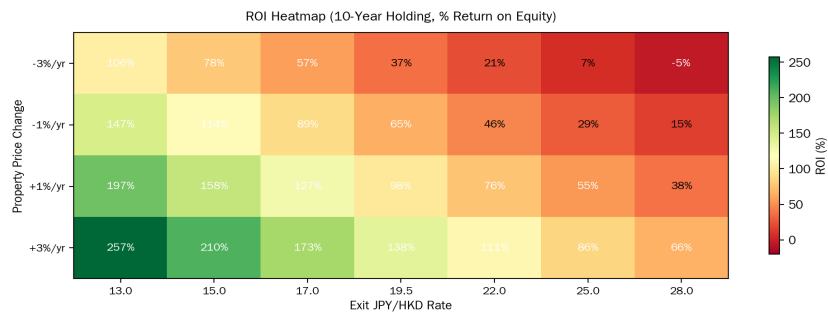
4.3 Holding: 10 Years

Prop \ FX	13.0	15.0	17.0	19.5	22.0	25.0	28.0
-3%/yr	+106% +203k	+78% +150k	+57% +110k	+37% +71k	+21% +41k	+7% +13k	-5% -9k
-1%/yr	+147% +283k	+114% +219k	+89% +171k	+65% +125k	+46% +89k	+29% +55k	+15% +28k
+1%/yr	+197% +379k	+158% +303k	+127% +245k	+98% +189k	+76% +145k	+55% +105k	+38% +73k

+3%/yr	+257% +494k	+218% +402k	+173% +332k	+138% +265k	+111% +213k	+86% +165k	+66% +126k
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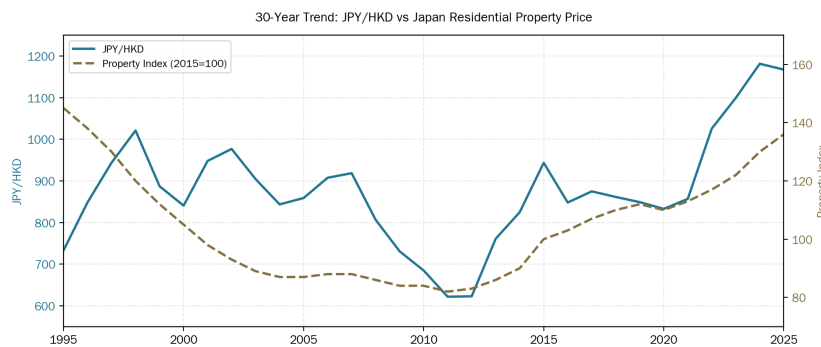
4.4 Visual: 10-Year ROI Heatmap

Exchange rate (horizontal) dominates the outcome. Even with falling property prices, a weaker JPY at exit can produce strong positive returns. The red zone (top-left) only occurs when JPY strengthens significantly AND property falls simultaneously - a rare combination historically.



5. Historical Evidence: 30-Year Summary

We analyzed every rolling 10-year window from 1995 to 2025 (16 periods total) to identify the actual worst, average, and best outcomes that historically occurred. By applying those exact FX and property changes to our model, we can show what your HKD 1.92M investment would have returned under real historical conditions.

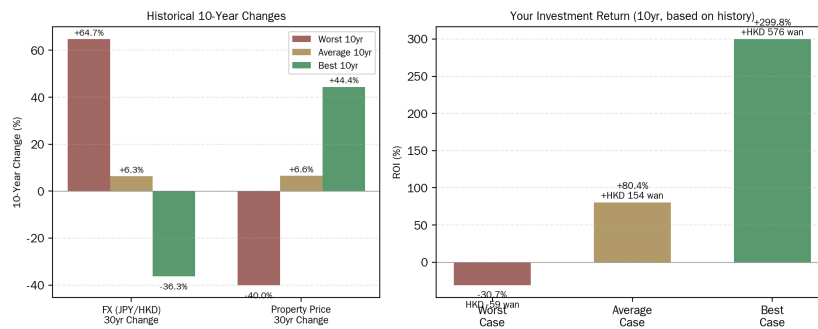


5.1 30-Year Cumulative Changes

Over 30 years, JPY/HKD rose from 733 to 1,167 (+59.2%), meaning JPY weakened substantially against HKD. For HKD investors, this long-term JPY weakening trend is a powerful tailwind: when you sell Japanese assets and convert back to HKD, each yen is worth more HKD than when you invested. Property prices nationally declined -6.2% over the same period, but urban Osaka/Tokyo significantly outperformed the national average, with cumulative gains of 30-40% in prime areas since 2012.

5.2 Historical 10-Year Scenarios: Visual Summary

The left chart shows the actual 10-year FX and property changes from historical data (worst, average, best). The right chart translates those changes into your specific investment return, measured in both ROI% and HKD.



5.3 Your Investment: 3 Historical Scenarios (10-Year Holding)

The table below applies the exact historical 10-year changes to your HKD 1.92M equity investment. This shows what you would have actually earned or lost if you had invested under those conditions.

	Worst Case	Average Case	Best Case
Historical Period (FX)	2012-2022	Average of 16 periods	2002-2012
Historical Period (Property)	1995-2005	Average of 16 periods	2014-2024
FX 10-Year Change	+64.8%	+6.3%	-36.3%
Property 10-Year Change	-40.0%	+6.6%	+44.4%
Exit Property Value	JPY 37.4M	JPY 66.5M	JPY 90.1M
Outstanding Loan at Exit	JPY 9.6M	JPY 9.6M	JPY 9.6M
Net Exit Proceeds (JPY)	JPY 27.8M	JPY 56.9M	JPY 80.5M

Accumulated Rental (10yr)	JPY 14.9M	JPY 14.9M	JPY 14.9M
Total Return (HKD)	HKD 133.0M	HKD 346.3M	HKD 767.7M
ROI on Equity	-30.7%	+80.3%	+299.8%
HKD Gain / Loss	-59 wan	+154 wan	+576 wan

Key insight: Even in the worst historical 10-year period (JPY weakening 64.7% + property falling 40.0%), the loss is -30.7% (-HKD 59 wan), cushioned by JPY 14.9M accumulated positive rental income. The average scenario returns +80.3% (+HKD 154 wan), nearly doubling your equity. The best scenario delivers +299.8% (+HKD 576 wan), a 4x return on your original investment.

6. Risk Scorecard

Statistical Summary (All 84 Scenarios)

Bottom Line

93% of 84 scenarios produce positive returns. Historical average scenario: +80.3% ROI. Worst historical 10-year: -30.7% (contained by positive cashflow). The key risk is short-term JPY strength from BOJ tightening; investors with 7-10 year horizons are well-positioned to capture rental income and FX tailwinds.